📊 SQL ANALYTICS REPORT: SUPERSTORE PROJECT

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Technology: MySQL Workbench

Scope: 8 Key Business Questions

1. 🎯 ANALYTICAL OBJECTIVE

The goal was to audit the Superstore dataset using MySQL to validate customer behavior, sales trends, and operational efficiency before visualizing the data in Power BI.

2. 🛠️ QUERY BREAKDOWN & BUSINESS INSIGHTS

Q1: Profitability & Loss Analysis.

Objective: Identify sub-categories that are bleeding money (High Loss %).

SELECT category, sub_category,

*ROUND((SUM(CASE WHEN Profit < 0 THEN 1 ELSE 0 END) / COUNT(*))*100, 2) AS Loss_Percentage*

FROM superstore_cleaned

GROUP BY category, sub_category

ORDER BY Loss_Percentage DESC;

Insight: "Tables" and "Bookcases" have the highest frequency of loss-making transactions, suggesting a need for immediate pricing review.

Q2: Time-Series Trend Analysis.

Objective: Track monthly sales performance.

SELECT YEAR(order_date), MONTH(order_date), ROUND(SUM(sales), 2) AS Total_Sales

FROM superstore_cleaned

GROUP BY YEAR(order_date), MONTH(order_date);

Insight: Validated that Q4 (Oct-Dec) is the peak revenue season.

Q3: Product Extremes (Best Seller).

Objective: Find the single most profitable product.

SELECT product_name, ROUND(SUM(profit)) AS Total_Profit

FROM superstore_cleaned

GROUP BY product_name

ORDER BY Total_Profit DESC LIMIT 1;

Insight: The Canon ImageCLASS Copier is the undisputed profit leader.

Q4: Most Frequent Customer (Loyalty).

Objective: Identify the customer with the highest number of visits/orders.

```
SELECT customer_name, COUNT(customer_name) AS Total_Visit
FROM superstore_cleaned
GROUP BY customer_name
ORDER BY Total_Visit DESC LIMIT 1;
```

Insight: Identifies the most loyal customer who visits frequently, even if their basket size is small. Use this for "Frequency Reward Programs".

Q5: Highest Revenue Customer (Whale).

Objective: Identify the customer who has spent the most money (Total Sales).

```
SELECT customer_name, ROUND(SUM(sales)) AS Total_Sales
FROM superstore_cleaned
GROUP BY customer_name
ORDER BY Total_Sales DESC LIMIT 1;
```

Insight: Differentiates high-value buyers (Whales) from frequent buyers. Sean Miller often tops this list, aligning with the Power BI dashboard.

Q6: Regional Champions (Window Functions).

Objective: Find the top customer per region using ranking.

```
 WITH Sal AS (
   SELECT region, customer_name, ROUND(SUM(sales), 2) AS Total_Sales,
         DENSE_RANK() OVER (PARTITION BY region ORDER BY SUM(sales) DESC) AS
Rnk
   FROM superstore_cleaned
   GROUP BY region, customer_name
)
```

*SELECT * FROM Sal WHERE Rnk = 1;*

Insight: Verified the list of Regional Leaders (Sean, Tamara, Tom, Raymond) used to fix the Power BI context error.

Q7: Year-over-Year (YoY) Growth.

Objective: Calculate annual growth rate using LAG.

SELECT order_year, current_sale,

LAG(current_sale) OVER(ORDER BY order_year) AS previous_sale,

*ROUND(((current_sale - LAG(current_sale) OVER(ORDER BY order_year)) / LAG(current_sale) OVER(ORDER BY order_year)) * 100, 2) AS Growth_Rate*

FROM yrsale;

Insight: Tracks the business scaling trajectory.

Q8: Delivery Performance (Logistics Audit).

Objective: Calculate average shipping time per mode.

SELECT ship_mode,

ROUND(AVG(DATEDIFF(ship_date, order_date)), 2) AS avg_delivery_days

FROM superstore_cleaned

GROUP BY ship_mode;

Insight:

Standard Class: ~5 Days

Same Day: ~0 Days (Instant)

Conclusion: Logistics are meeting SLAs.